

Public Report – Project AtendeAI

Module 3: Application Integration with Image and Voice Features

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1. Module Overview

This module represents a significant milestone in achieving full application integration. By implementing image support, **Text-to-Speech (TTS)**, and **Speech-to-Text (STT)**, we have introduced critical accessibility features that enhance the user experience. Furthermore, our research into AI bias and development costs provided essential insights that have shaped the project's workflow, underscored its importance, and guided our financial decision-making.

2. Multimodal Cloud Application

2.1 Image

The **image processing module** enables users to send photos or screenshots as part of their service requests.

When an image is uploaded, the system routes it through a **computer vision agent**, hosted on a cloud instance and connected to the main backend.

This agent applies **object and text detection (OCR)** models to identify relevant details such as documents, street elements, or damage reports. The extracted data is then associated with the user's request in the MongoDB database, allowing automated tagging and faster triage by municipal services.

This multimodal capacity increases accessibility for citizens who prefer sending visual evidence rather than typing detailed descriptions.

2.2 STT

This application implements **Speech-to-Text (STT)** functionality to convert spoken audio into text. The process begins on the **React frontend**, where the user's speech is recorded and sent as an audio file (Blob) to the **NodeJS backend**.

The backend acts as a secure intermediary, forwarding the audio to the **IBM Watson STT API**. This AI service processes the audio and returns the transcribed text, which the backend then sends back to the frontend for display. This implementation significantly enhances the user experience by offering greater accessibility, efficiency, and a modern, competitive edge to the application.

2.3 TTS

The **Text-to-Speech (TTS)** module converts system messages and chatbot responses into audio, enhancing accessibility for visually impaired users and enabling voice interactions.

It integrates with **cloud-based voice synthesis APIs** such as IBM Watson TTS or Google Cloud Text-to-Speech, processing up to **2 million characters per month**, as estimated in the cost analysis.

The audio output is streamed back to the frontend, where users can listen directly in the browser or app.

This feature supports only **Portuguese**, but it contributes to a more inclusive user experience and compliance with accessibility guidelines.

3. Biases in Artificial Intelligence

Biases in Artificial Intelligence can be understood as distortions or systematic tendencies that lead a model to produce unfair or unbalanced results regarding specific groups or situations. These biases may arise from various sources, such as non-representative training data, model design choices, or even misinterpretations of results.

In the AtendeAI project, whose purpose is to support citizens and public services through automated image analysis, understanding and mitigating biases is crucial to ensuring that the solution is inclusive, fair, and reliable. If not properly addressed, a bias could compromise the correct classification of a citizen's report, reducing the efficiency of the service.

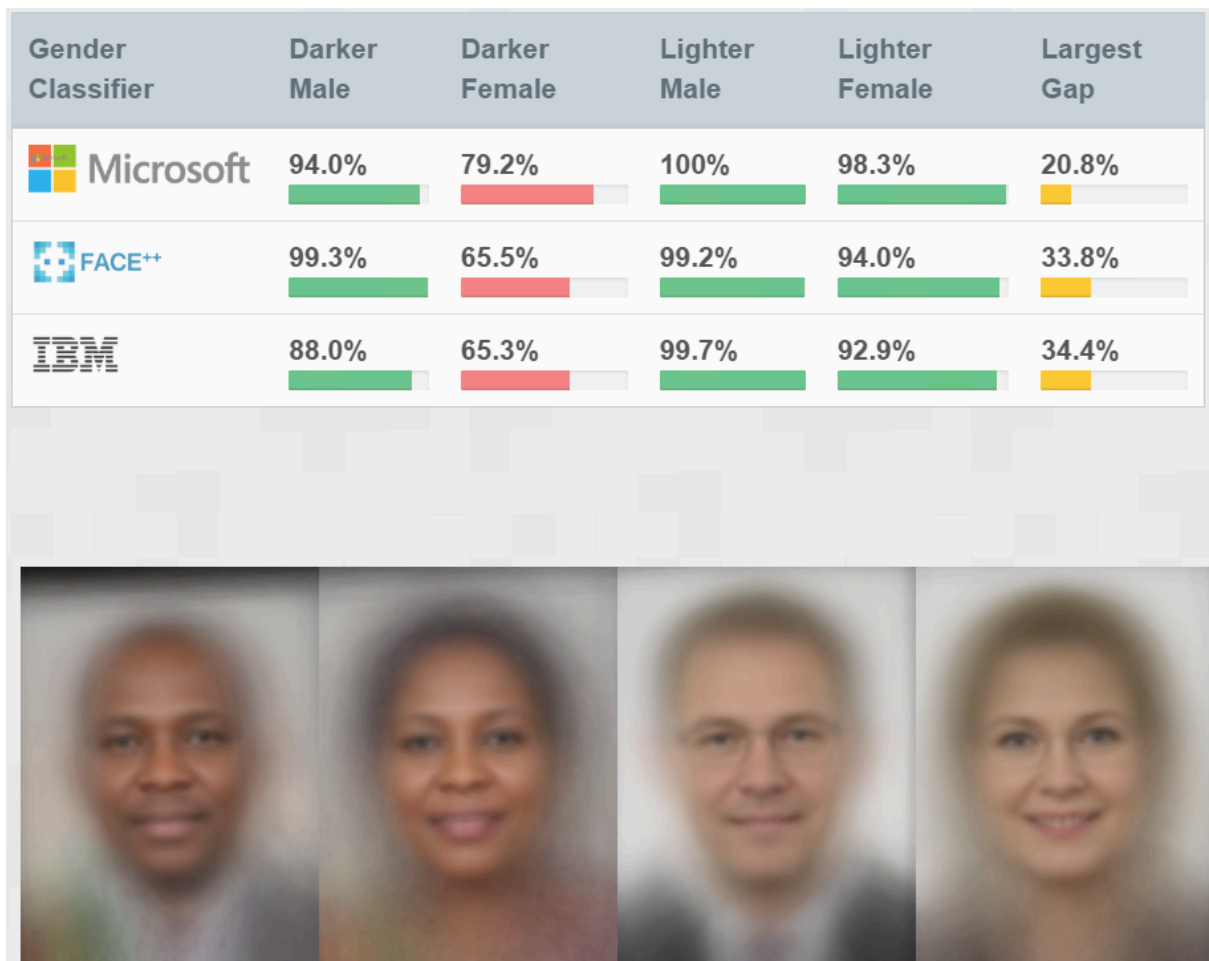
3.1 Application of Biases in Visual Models

In image recognition and visual tagging systems, biases may appear in different ways. Examples include:

- Facial recognition: models trained mostly on white faces may show lower accuracy for Black or Asian individuals.
- Object classification: datasets dominated by urban contexts may struggle to correctly identify objects in rural areas.
- Specific situations: if training images prioritize certain scenarios (e.g., trash in downtown areas), the model may fail to correctly identify the same issue in peripheral neighborhoods.

A notorious case occurred in 2015, when Google's photo system mistakenly labeled Black people as "gorillas." This error was not only technical but also a direct reflection of biased training data, and it raised major public concern about the social risks of AI.

The Gender Shades Project, led by Joy Buolamwini and Timnit Gebru (2018), was key in scientifically exposing inequalities in the performance of commercial facial recognition systems. The study evaluated models from major companies (IBM, Microsoft, and Face++) and found that error rates were below 1% for light-skinned men, but could reach 34% for dark-skinned women. This result proved that datasets were biased, favoring male and white faces, and highlighted the urgent need for more diverse training data.



The impact of biases in AI goes far beyond the technical sphere. In real-world contexts, especially in public services, biases can generate inequalities and reinforce social stigmas. For instance, if AtendeAI fails to correctly classify images submitted by citizens from certain regions or social groups, this could result in inefficient or discriminatory services.

From an ethical perspective, this raises issues related to:

- Equity: all citizens should receive the same level of service.
- Transparency: it is important that the model's limitations and potential biases are clearly communicated.
- Social responsibility: systems applied in public policies can directly affect people's lives, making impartiality a critical requirement.

These reflections show that technology must be accompanied by processes such as continuous auditing, use of diverse datasets, and multidisciplinary expert evaluation.

4. Similar Solutions Comparison in the Market and Cost Estimate

This report presents a comparative analysis between the proposed AtendeAI solution and existing public digital service platforms. It includes methodology, an overview of selected solutions, comparative data tables, critical analysis, and cost estimations.

Note: All comparisons are based on publicly available information.

4.1. Methodology

Sources: Official government portals, public documentation, and institutional announcements.

Criteria: Features, service channels, automation level, architecture, scalability, UX, and transparency.

Note: Some data is limited due to restricted public access.

4.2. Solutions Survey

SP156 — São Paulo City Hall

- Platform for urban service requests (lighting, maintenance, etc.).
- Channels: Phone (156), website, mobile app.
- Features: Protocol tracking, image upload, integration with city departments.
- Remarks: High service demand; low automation; form-based UX.

Conecta Recife

- Municipal portal for local service management and scheduling.
- Channels: Mobile app, website, and in-person service points.
- Features: Appointments, document uploads, protocol tracking.
- Remarks: Effective local integration; limited automation and AI usage.

Gov.br — National Portal

- Federal digital platform with unified citizen access and authentication.
- Investment: BRL 43 million (2020) for development, hosting, and migration.
- Savings: Estimated BRL 100 million per year from centralizing 1,600 websites.
- Channels: Web, app, integrated APIs.
- Remarks: Large-scale solution; experimental AI; variable UX.

AtendeAI — Proposed Solution

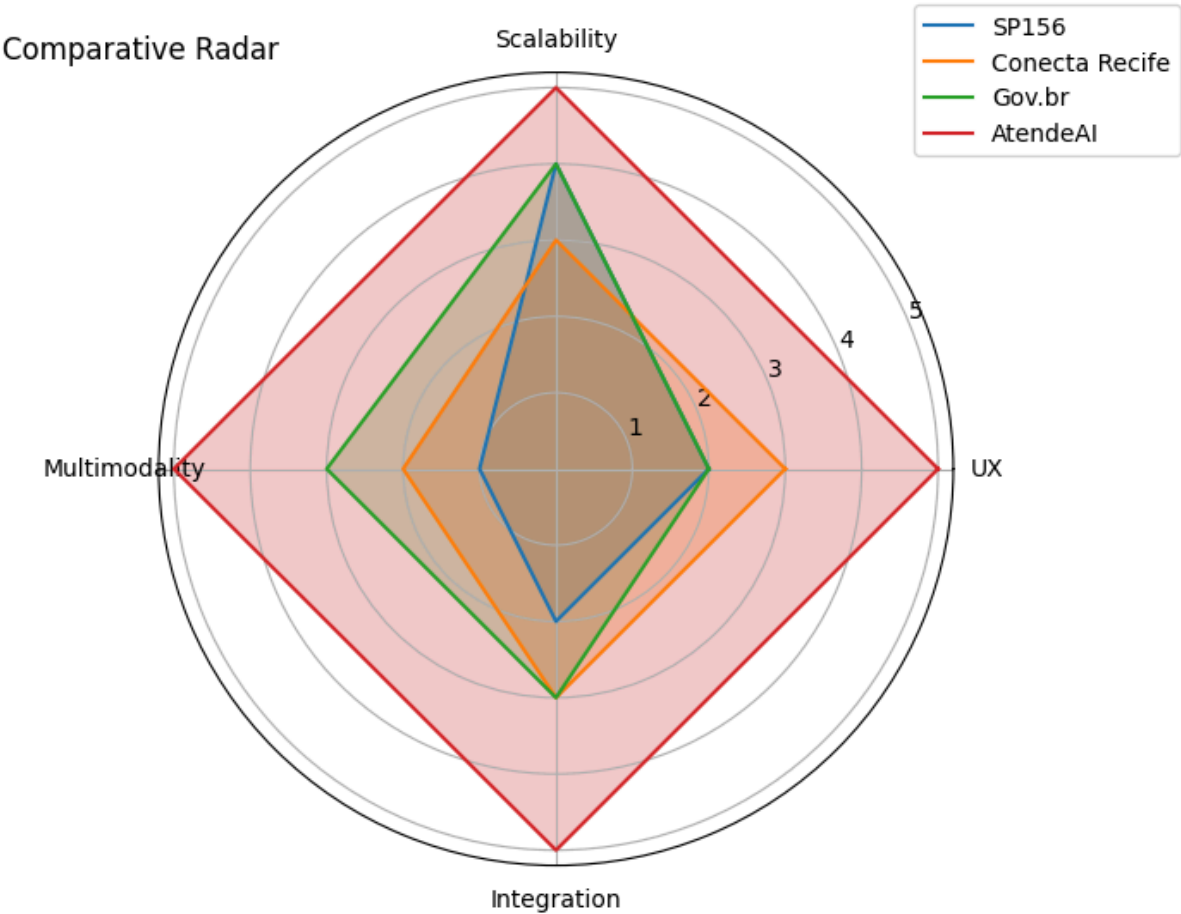
- Cloud-native, AI-driven public service assistant.
- Features: NLP/NLU, speech recognition, OCR, computer vision, multimodal inputs (text, voice, image, video).
- Channels: Web, app, WhatsApp, API, voice assistant.
- Remarks: Conversational interface; scalable cloud infrastructure; focus on accessibility and automation.

4.3. Comparative Overview

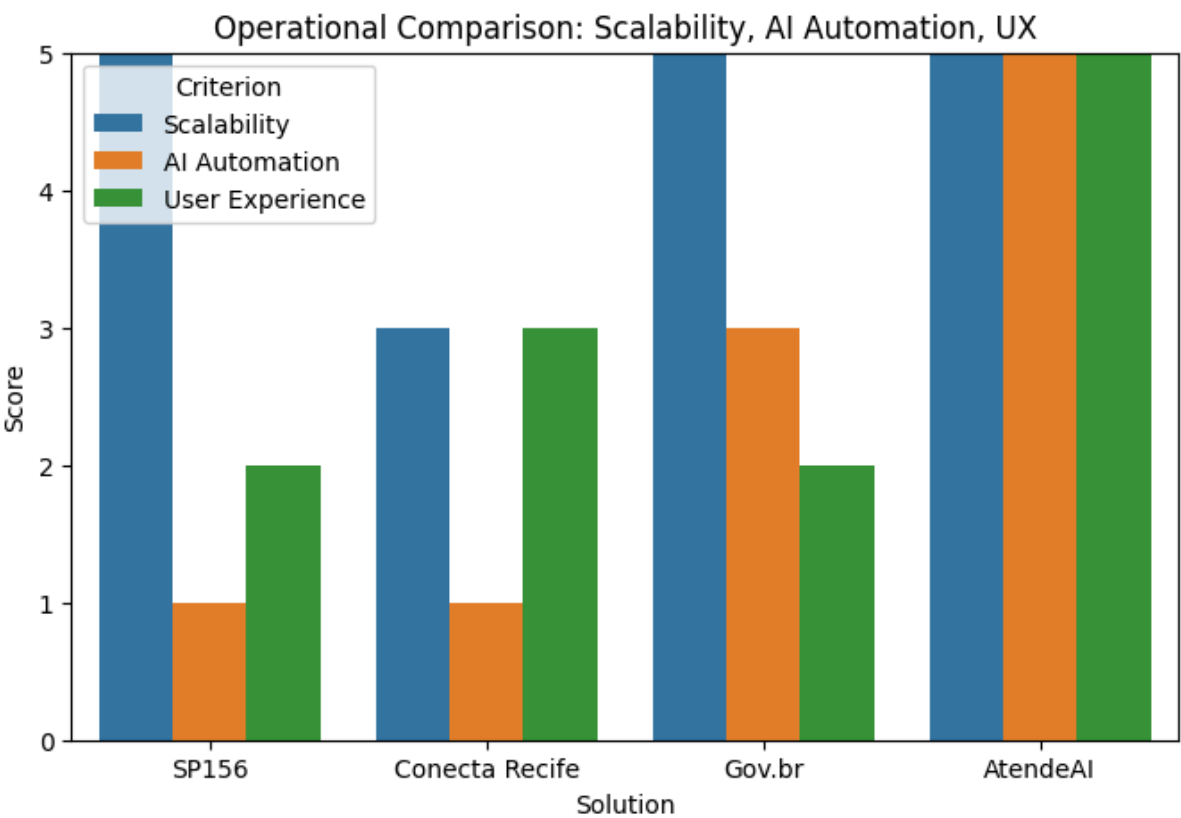
Feature Comparison Table

Criterion	SP156	Conecta Recife	Gov.br	AtendeAI (Proposed)
Input	Text, phone, image	Text, image/document	Text, document, chatbot	Text, voice, image, video
AI / Automation	Low	Low	Medium	High
Channels	Phone, web, app	App, web, in-person	Web, app, API	Web, app, WhatsApp, API, voice
Architecture	Municipal	Local	Federal	Cloud-native, API-first
Scalability	High	Medium	High	Auto-scaling cloud
UX	Bureaucratic	Simplified	Variable	Conversational
Transparency	Public protocols	Local protocols	Gov.br authentication	Auto protocol tracking

Comparative Radar Graph



Operational Comparison Graph



Infrastructure & Cost Overview

Aspect	SP156	Conecta Recife	Gov.br	AtendeAI (Proposed)
Infrastructure	City servers	Local servers	Federal infrastructure	Cloud (AWS/Azure/GCP)
AI Integration	Minimal	Minimal	Experimental	NLP, OCR, ASR integrated
Team Structure	City Hall	Municipal IT	Federal ministry teams	Small DevOps + analysts
Cost Profile	Medium/High	Medium	BRL 43M initial / BRL 100M yearly savings	Scalable by usage

4.4. Critical Analysis

- SP156: Strong local service coverage, limited automation.
- Conecta Recife: Centralized but lacks innovation and scalability.
- Gov.br: Impressive scope and savings potential but inconsistent UX.
- AtendeAI: Superior automation, integration, and scalability.

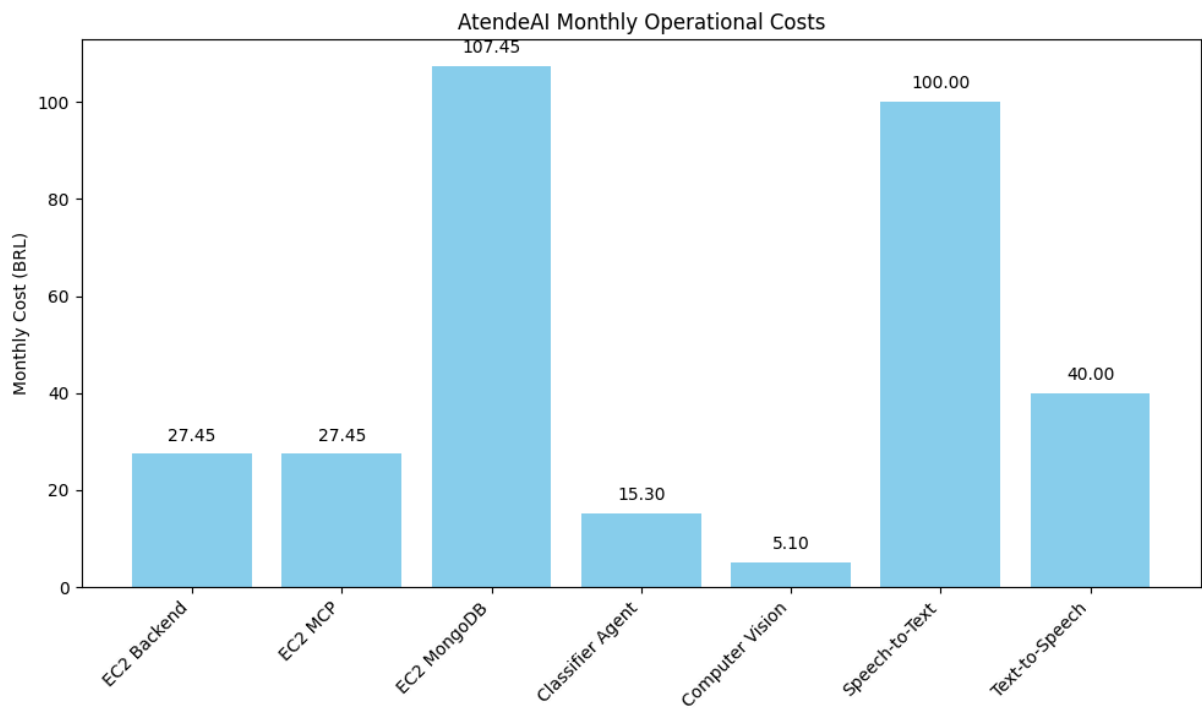
4.5. Operational Cost Estimate — AtendeAI

Component	Monthly Cost (BRL)
EC2 – Backend	27.45
EC2 – MCP	27.45
EC2 – MongoDB	107.45
Classifier Agent	15.30
Computer Vision Agent	5.10
Speech-to-Text (5k min/month)	100.00
Text-to-Speech (2M chars/month)	40.00

Total Estimated Monthly Cost: BRL 322.75

Notes: Cloud components scale dynamically with usage. Costs reflect realistic SaaS consumption. Future versions may integrate optimization strategies and multi-cloud setups.

Monthly Operational Costs Bar Graph



5. Final Considerations

The comparative study demonstrated that while current public service platforms such as **SP156**, **Conecta Recife**, and **Gov.br** offer strong infrastructures, they still face challenges in automation, accessibility, and multimodality.

The **AtendeAI** proposal stands out by integrating **text, voice, and image inputs** under a **cloud-native and API-first** design, resulting in scalable and user-friendly interactions.

Operational cost simulations confirm that this model is **financially viable**, with moderate monthly expenses and elastic scalability according to usage. The project emphasizes how **artificial intelligence can modernize public service access**, reduce response times, and improve user satisfaction. Future steps include expanding dataset diversity, refining conversational accuracy, and exploring integration with real municipal databases.

6. References

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